

Answer 1-

```
akshansh@vbox: ~  
File Actions Edit View Help  
--(akshansh@vbox)-[~]  
$ ip link show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000  
    link/ether 08:00:27:18:9d:21 brd ff:ff:ff:ff:ff:ff  
--(akshansh@vbox)-[~]  
$ sudo ifconfig eth0 up;sudo dumpcap -i eth0 -c 250 -w ~/temp/sample1.pcap  
Capturing on 'eth0'  
dumpcap: The file to which the capture would be saved ("/home/akshansh/temp/sample1.pcap") could not be opened: Permission denied.  
--(akshansh@vbox)-[~]  
$
```

Answer 2-

```
akshansh@vbox: ~  
File Actions Edit View Help  
--(akshansh@vbox)-[~]  
$ sudo apt update  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/main arm64 Packages [20.5 MB]  
Get:3 http://kali.download/kali kali-rolling/main arm64 Contents (deb) [48.7 MB]  
Get:4 http://kali.download/kali kali-rolling/contrib arm64 Packages [104 kB]  
Get:5 http://kali.download/kali kali-rolling/contrib arm64 Contents (deb) [192 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free arm64 Packages [149 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free arm64 Contents (deb) [859 kB]  
Get:8 http://kali.download/kali kali-rolling/non-free-firmware arm64 Packages [11.0 kB]  
Get:9 http://kali.download/kali kali-rolling/non-free-firmware arm64 Contents (deb) [29.0 kB]  
Fetched 70.6 MB in 1min 59s (595 kB/s)  
2220 packages can be upgraded. Run 'apt list --upgradable' to see them.  
--(akshansh@vbox)-[~]  
$ sudo apt install tinyproxy  
The following packages were automatically installed and are no longer required:  
  gir1.2-girepository-2.0 libgirepository-1.0-1  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  tinyproxy  
  
Installing dependencies:  
  tinyproxy-bin  
  
Summary:  
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2220  
  Download size: 61.5 kB  
  Space needed: 240 kB / 7,644 MB available  
  
Continue? [Y/n] y  
Get:1 http://kali.kali.org/kali kali-rolling/main arm64 tinyproxy-bin arm64 1.11.2-1+b2 [41.6 kB]  
Get:2 http://kali.download/kali kali-rolling/main arm64 tinyproxy all 1.11.2-1 [19.9 kB]  
Fetched 61.5 kB in 2s (26.6 kB/s)  
Selecting previously unselected package tinyproxy-bin.  
(Reading database ... 406615 files and directories currently installed.)  
Preparing to unpack .../tinyproxy-bin_1.11.2-1+b2_arm64.deb ...  
Unpacking tinyproxy-bin (1.11.2-1+b2) ...  
Selecting previously unselected package tinyproxy.  
Preparing to unpack .../tinyproxy_1.11.2-1_all.deb ...  
Unpacking tinyproxy (1.11.2-1) ...  
Setting up tinyproxy-bin (1.11.2-1+b2) ...
```

```
File Actions Edit View Help
Fetched 61.5 kB in 2s (26.6 kB/s)
Selecting previously unselected package tinyproxy-bin.
(Reading database ... 406615 files and directories currently installed.)
Preparing to unpack .../tinyproxy-bin_1.11.2-1+b2_arm64.deb ...
Unpacking tinyproxy-bin (1.11.2-1+b2) ...
Selecting previously unselected package tinyproxy.
Preparing to unpack .../tinyproxy_1.11.2-1_all.deb ...
Unpacking tinyproxy (1.11.2-1) ...
Setting up tinyproxy-bin (1.11.2-1+b2) ...
Setting up tinyproxy (1.11.2-1) ...
update-rc.d: As per Kali policy, tinyproxy init script is left disabled.
tinyproxy.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.2.7) ...

(akshansh@vbox)~$ sudo vi /etc/tinyproxy.conf
sudo: vi/etc/tinyproxy.conf: command not found

(akshansh@vbox)~$ sudo /etc/init.d/tinyproxy stop
Stopping tinyproxy (via systemctl): tinyproxy.service.

(akshansh@vbox)~$ sudo apt remove tinyproxy
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0 libgirepository-1.0-1 tinyproxy-bin
Use 'sudo apt autoremove' to remove them.

REMOVING:
  tinyproxy

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 2220
  Freed space: 79.9 kB

Continue? [Y/n] y
(Reading database ... 406647 files and directories currently installed.)
Removing tinyproxy (1.11.2-1) ...
Processing triggers for man-db (2.13.1-1) ...

(akshansh@vbox)~$
```

Feb 8 16:07

3.1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.090600000	fd17:625c:f037:2:c6...	2620:2d:4000:1::22	TCP	94	59082 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 S
2	2.040813762	fd17:625c:f037:2:c6...	2620:2d:4000:1::22	TCP	94	[TCP Retransmission] 59082 → 80 [SYN] Seq=0 Win=6
3	6.075897948	fd17:625c:f037:2:c6...	2620:2d:4000:1::22	TCP	94	[TCP Retransmission] 59082 → 80 [SYN] Seq=0 Win=6
4	12.763112868	2620:2d:4000:1::22	fd17:625c:f037:2:c6...	TCP	74	80 → 59082 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
5	117.718997351	10.0.2.15	172.236.180.15	NTP	90	NTP Version 4, client
6	117.863135133	172.236.180.15	10.0.2.15	NTP	90	NTP Version 4, server
7	122.361431343	10.0.2.15	10.0.2.3	DNS	89	Standard query 0x9b8a AAAA connectivity-check.ubu
8	122.466238030	10.0.2.3	10.0.2.15	DNS	425	Standard query response 0x9b8a AAAA connectivity-
9	122.467819088	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	55544 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 S
10	122.812106468	PCSSystemtec_64:4b...	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
11	122.812497484	52:55:0a:00:02:02	PCSSystemtec_64:4b...	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02
12	122.988153250	10.0.2.15	224.0.0.251	MDNS	87	Standard query 0x6000 PTR _ipp.tcp.local, "QM" q
13	122.988357240	fd17:625c:f037:2:a0...	ff02::fb	MDNS	107	Standard query 0x6000 PTR _ipps.tcp.local, "QM" q
14	123.452518385	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=6
15	124.471990522	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=6

Frame 1: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface 0

Ethernet II, Src: PCSSystemtec_64:4b:d1 (08:00:27:00:00:00), Dst: 08:00:00:00:00:00

Internet Protocol Version 6, Src: fd17:625c:f037:2:c6::1, Dst: 2620:2d:4000:1::22

Transmission Control Protocol, Src Port: 59082, Dst Port: 80

0000 52 56 00 00 00 02 08 00 27 64 4b d1 86 dd 60 0c RV.....'dK....

0010 7a d8 00 28 06 40 fd 17 62 5c f0 37 00 02 c6 52 z: (. @ . . b \ . 7 . . R

0020 29 44 84 13 1e 50 26 20 00 2d 40 00 00 01 60 00) D . . P & . . @

0030 00 00 00 00 00 00 22 e6 ca 00 50 e8 cb 17 d4 60 00 " . . P

0040 00 00 a0 02 fd 20 48 47 00 00 02 04 65 a0 64 02 HG

0050 08 0a fc 1a 2c d5 00 00 00 00 01 03 03 07 h . y

3.1.pcapng

Packets: 9254 · Displayed: 9254 (100.0%) Profile: Default

3.1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fd17:625c:f037:2:c6...	2620:2d:4000:1::22	TCP	94	59082 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 S
2	2.049813752	fd17:625c:f037:2:c6...	2620:2d:4000:1::22	TCP	94	[TCP Retransmission] 59082 → 80 [SYN] Seq=0 Win=0
3	6.075907948	fd17:625c:f037:2:c6...	2620:2d:4000:1::22	TCP	94	[TCP Retransmission] 59082 → 80 [SYN] Seq=0 Win=0
4	12.703112868	2620:2d:4000:1::22	fd17:625c:f037:2:c6...	TCP	74	80 → 59082 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
5	117.718997351	10.0.2.15	172.236.180.15	NTP	96	NTP Version 4, client
6	117.803135133	172.236.180.15	10.0.2.15	NTP	96	NTP Version 4, server
7	122.361431343	10.0.2.15	10.0.2.3	DNS	89	Standard query 0x9b8a AAAA connectivity-check.ubu
8	122.406238036	10.0.2.3	10.0.2.15	DNS	425	Standard query response 0x9b8a AAAA connectivity-
9	122.407819088	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	55544 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 S
10	122.812105408	PCSSystemtec_64:4b:...	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
11	122.812497464	52:55:0a:00:02:02	PCSSystemtec_64:4b:...	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02
12	122.988153250	10.0.2.15	224.0.0.251	MDNS	87	Standard query 0x6000 PTR _ipp._tcp.local, "QM" q
13	122.988357248	fd17:625c:f037:2:a0...	ff02::fb	MDNS	107	Standard query 0x6000 PTR _ipps._tcp.local, "QM" q
14	123.452518385	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=0
15	124.471999522	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=0

Frame 1: 94 bytes on wire (752 bits), 94 bytes captured on interface 0, 94 bytes from 2620:2d:4000:1::22 to fd17:625c:f037:2:c6... on interface 0

Ethernet II, Src: PCSSystemtec_64:4b:d1 (08:00:27:00:00:00), Dst: fd17:625c:f037:2:c6... (00:00:00:00:00:00)

Internet Protocol Version 6, Src: fd17:625c:f037:2:c6..., Dst: 2620:2d:4000:1::22

Transmission Control Protocol, Src Port: 59082, Dst Port: 80

3.1.pcapng

Packets: 9254 · Displayed: 9254 (100.0%) Profile: Default

3.1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
19	127.419424783	PCSSystemtec_64:4b:...	52:55:0a:00:02:03	ARP	42	Who has 10.0.2.3? Tell 10.0.2.15
20	127.419623149	fe80::2	fe80::a00:27ff:fe64::	ICMPv6	86	Neighbor Advertisement fe80::2 (rtr, sol, ovr) is
21	127.419900208	52:55:0a:00:02:03	PCSSystemtec_64:4b:...	ARP	64	10.0.2.3 is at 52:55:0a:00:02:03
22	127.736087602	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=0
23	129.785473153	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=0
24	133.817921147	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=0
25	142.265810952	fd17:625c:f037:2:c6...	2620:2d:4002:1::196	TCP	94	[TCP Retransmission] 55544 → 80 [SYN] Seq=0 Win=0
26	143.410110309	2620:2d:4002:1::196	fd17:625c:f037:2:c6...	TCP	74	80 → 55544 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
27	163.880549771	10.0.2.15	10.0.2.3	DNS	72	Standard query 0x9ef6 A ubuntu.Dlink
28	163.989722573	10.0.2.3	10.0.2.15	DNS	147	Standard query response 0x9ef6 No such name A ubu
29	164.016355831	10.0.2.15	10.0.2.3	DNS	72	Standard query 0xc0f7 A ubuntu.Dlink
30	164.027211138	10.0.2.3	10.0.2.15	DNS	72	Standard query response 0xc0f7 No such name A ubu
31	164.703332462	10.0.2.15	10.0.2.3	DNS	89	Standard query 0x2bd5 A connectivity-check.ubuntu
32	164.728732715	10.0.2.3	10.0.2.15	DNS	281	Standard query response 0x2bd5 A connectivity-che
33	164.730118950	10.0.2.15	91.189.91.49	TCP	74	36120 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S

Frame 1: 94 bytes on wire (752 bits), 94 bytes captured on interface 0, 94 bytes from 2620:2d:4000:1::22 to fd17:625c:f037:2:c6... on interface 0

Ethernet II, Src: PCSSystemtec_64:4b:d1 (08:00:27:00:00:00), Dst: fd17:625c:f037:2:c6... (00:00:00:00:00:00)

Internet Protocol Version 6, Src: fd17:625c:f037:2:c6..., Dst: 2620:2d:4000:1::22

Transmission Control Protocol, Src Port: 59082, Dst Port: 80

3.1.pcapng

Packets: 9254 · Displayed: 9254 (100.0%) Profile: Default

3.1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
91	168.024811094	10.0.2.15	10.0.2.3	DNS	84	Standard query 0x2be4 AAAA detectportal.firefox.c
92	168.039341148	10.0.2.3	10.0.2.15	DNS	195	Standard query response 0xec9d A detectportal.fir
93	168.039809288	10.0.2.3	10.0.2.15	DNS	207	Standard query response 0x2be4 AAAA detectportal.
94	168.085443088	34.160.144.191	10.0.2.15	TLSv1.2	2934	Server Hello, Certificate
95	168.085495649	10.0.2.15	34.160.144.191	TCP	54	45396 → 443 [ACK] Seq=221 Ack=2881 Win=65535 Len=
96	168.085659915	34.160.144.191	10.0.2.15	TLSv1.2	1494	Server Hello
97	168.085678781	10.0.2.15	34.160.144.191	TCP	54	45408 → 443 [ACK] Seq=221 Ack=1441 Win=65535 Len=
98	168.086233827	34.160.144.191	10.0.2.15	TLSv1.2	202	Server Key Exchange, Server Hello Done
99	168.086234078	34.160.144.191	10.0.2.15	TLSv1.2	1494	Certificate
100	168.086234168	34.160.144.191	10.0.2.15	TLSv1.2	202	Server Key Exchange, Server Hello Done
101	168.086258354	10.0.2.15	34.160.144.191	TCP	54	45396 → 443 [ACK] Seq=221 Ack=3029 Win=65535 Len=
102	168.086275708	10.0.2.15	34.160.144.191	TCP	54	45408 → 443 [ACK] Seq=221 Ack=2881 Win=65535 Len=
103	168.086284986	10.0.2.15	34.160.144.191	TCP	54	45408 → 443 [ACK] Seq=221 Ack=3029 Win=65535 Len=
104	168.087429384	10.0.2.15	34.160.144.191	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypte
105	168.087707558	34.160.144.191	10.0.2.15	TCP	60	443 → 45396 [ACK] Seq=3029 Ack=314 Win=65535 Len=

Frame 1: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface 0
 Ethernet II, Src: PCSSystemtec_64:4b:d1 (08:00:27:64:4b:d1), Dst: 08:00:27:64:4b:d1
 Internet Protocol Version 6, Src: fd17:625c:f037:2::1, Dst: fd17:625c:f037:2::3
 Transmission Control Protocol, Src Port: 59082, Dst Port: 443

3.1.pcapng Packets: 9254 · Displayed: 9254 (100.0%) Profile: Default

3.1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
124	168.189855509	34.160.144.191	10.0.2.15	TCP	60	443 → 45396 [ACK] Seq=3409 Ack=760 Win=55535 Len=
125	168.189945625	10.0.2.15	185.125.188.59	TCP	54	55232 → 443 [ACK] Seq=9602 Ack=19537 Win=65535 Len=
126	168.190025776	34.160.144.191	10.0.2.15	TCP	60	443 → 45408 [ACK] Seq=3409 Ack=462 Win=55535 Len=
127	168.193978072	10.0.2.15	34.107.221.82	TCP	74	47114 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
128	168.198815212	34.160.144.191	10.0.2.15	TLSv1.2	92	Application Data
129	168.201713600	34.160.144.191	10.0.2.15	TLSv1.2	92	Application Data
130	168.201753447	10.0.2.15	34.160.144.191	TCP	54	45408 → 443 [RST] Seq=462 Win=0 Len=0
131	168.201904447	34.160.144.191	10.0.2.15	TCP	60	443 → 45408 [FIN, ACK] Seq=3447 Ack=462 Win=65535 Len=
132	168.201919395	10.0.2.15	34.160.144.191	TCP	54	45408 → 443 [RST] Seq=462 Win=0 Len=0
133	168.202008958	34.160.144.191	10.0.2.15	TCP	60	443 → 45408 [RST, ACK] Seq=4257335295 Ack=462 Win=
134	168.203626495	34.160.144.191	10.0.2.15	TLSv1.2	198	Application Data
135	168.204796952	34.107.221.82	10.0.2.15	TCP	60	80 → 47114 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
136	168.204855174	10.0.2.15	34.107.221.82	TCP	54	47114 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
137	168.211141947	10.0.2.15	34.160.144.191	TCP	54	45396 → 443 [ACK] Seq=760 Ack=3591 Win=55535 Len=
138	168.212816955	10.0.2.15	34.160.144.191	TLSv1.2	190	Application Data

Frame 1: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface 0
 Ethernet II, Src: PCSSystemtec_64:4b:d1 (08:00:27:64:4b:d1), Dst: 08:00:27:64:4b:d1
 Internet Protocol Version 6, Src: fd17:625c:f037:2::1, Dst: fd17:625c:f037:2::3
 Transmission Control Protocol, Src Port: 59082, Dst Port: 443

3.1.pcapng Packets: 9254 · Displayed: 9254 (100.0%) Profile: Default

